

LOGIC CURRICULUM

Unified Course Syllabus & Structure

Compiled Reference Manual

BLOCK 1: CLASSICAL LOGIC

Unit 1: Types of Categorical Propositions: A, E, I, O and Square of Opposition

- **1.0** Objectives
- **1.1** Introduction
- **1.2** Terms and their Kinds
- **1.3** Denotation and Connotation of Terms
- **1.4** Meaning and Supposition of Terms
- **1.5** Propositions
- **1.6** Square of Opposition
- **1.7** Let Us Sum Up
- **1.8** Key Words
- **1.9** Further Readings and References

Unit 2: Categorical Syllogism

- **2.0** Objectives
- **2.1** Introduction
- **2.2** Reason and Inference: Meaning and Objections
- **2.3** Kinds of Inference
- **2.4** Deductive Reasoning and Syllogism
- **2.5** Kinds of Syllogism
- **2.6** Let Us Sum Up
- **2.7** Key Words
- **2.8** Further Readings and References

Unit 3: Figure, Mood and the Possible Types of Syllogisms

- **3.0** Objectives
- **3.1** Introduction
- **3.2** Moods of Categorical Syllogism
- **3.3** Figures of Syllogism
- **3.4** Incomplete Syllogism and Compound Syllogism
- **3.5** Dilemma
- **3.6** Avoiding Dilemma
- **3.7** Let Us Sum Up
- **3.8** Key Words
- **3.9** Further Readings and References

Unit 4: Validity, Invalidity and List of Valid Syllogisms

- **4.0** Objectives
- **4.1** Introduction
- **4.2** The Rules of Categorical Syllogism
- **4.3** Special Applications of General Rules
- **4.4** Reduction of Arguments to I Figure
- **4.5** Antilogism or Inconsistent Triad
- **4.6** Venn Diagram Technique
- **4.7** Boolean Analysis
- **4.8** Let Us Sum Up
- **4.9** Key Words
- **4.10** Further Readings and References

BLOCK 2: SENTENTIAL LOGIC 1: INTRODUCTION

Unit 1: Introduction to the Form of Arguments in Modern Logic

- 1.0 Objectives
- 1.1 Introduction
- 1.2 A Short Story of Logic
- 1.3 Classical Logic and Symbolic Logic
- 1.4 Why use Symbols?
- 1.5 The Nature of Argument
- 1.6 Truth and Validity
- 1.7 Argument Forms
- 1.8 Truth-Table
- 1.9 Kinds of Sentence Forms and Sentences
- 1.10 Testing the Validity of Argument Forms
- 1.11 Exercises
- 1.12 Let Us Sum Up
- 1.13 Key Words
- 1.14 Further Readings and References

Unit 2: Conjunction, Disjunction, Conditional and Biconditional

- 2.0 Objectives
- 2.1 Introduction
- 2.2 Negation
- 2.3 Conjunction
- 2.4 Disjunction
- 2.5 Exercises
- 2.6 Implication
- 2.7 Biconditional
- 2.8 Let Us Sum Up
- 2.9 Key Words
- 2.10 Further Readings and References

Unit 3: Rules of Inference and the Nature of Validity of Arguments

- **3.0** Objectives
- **3.1** Introduction - Tools of Testing Arguments
- **3.2** Methods of Testing the Validity of Arguments
- **3.3** Application of Elementary Rules of Inference
- **3.4** Exercises
- **3.5** Let Us Sum Up
- **3.6** Key words
- **3.7** Further Readings and References

Unit 4: Fallacies

- **4.0** Objective
- **4.1** Introduction
- **4.2** Classification of Fallacies
- **4.3** Fallacies of Relevance
- **4.4** Fallacies of Induction
- **4.5** Fallacies of Presumption
- **4.6** Fallacies of Ambiguity
- **4.7** Exercises
- **4.8** Let Us Sum Up
- **4.9** Key Words
- **4.10** Further Readings and References

BLOCK 3: SENTENTIAL LOGIC 2: PROVING VALIDITY

Unit 1: Proving Validity Using Rules of Inference

- 1.0 Objectives
- 1.1 Introduction
- 1.2 Necessity of Rules of Inference
- 1.3 Meaning of Proof of Validity
- 1.4 Nine Rules of Inference
- 1.5 Usage of Rules of Inference to test Validity
- 1.6 Converting Verbal Forms of Argument into Symbols
- 1.7 Examples for Using Rules of Inference
- 1.8 Rules of Replacement
- 1.9 Let Us Sum Up
- 1.10 Key Words
- 1.11 Further Readings and References

Unit 2: Conditional Proof

- 2.0 Objectives
- 2.1 Introduction
- 2.2 Conditional Proof (C.P.)
- 2.3 Exercises I
- 2.4 The Strengthened Rule of C. P.
- 2.5 Exercises II
- 2.6 Let Us Sum Up
- 2.7 Key Words
- 2.8 Further Readings and References

Unit 3: Indirect Proof

- **3.0** Objectives
- **3.1** Introduction
- **3.2** The Meaning of Indirect Proof
- **3.3** Application of Indirect Proof
- **3.4** Examples
- **3.5** Exercises on Indirect Proof
- **3.6** Indirect Proof and Proof of Tautology
- **3.7** Let Us Sum Up
- **3.8** Key Words
- **3.9** Further Readings and References

Unit 4: Proving Invalidity

- **4.0** Objectives
- **4.1** Introduction
- **4.2** Proving Invalidity
- **4.3** Advantages of the Method of Proving Invalidity
- **4.4** Assumptions of Proving Invalidity
- **4.5** Second Method of Proving Invalidity; Examples
- **4.6** Exercises
- **4.7** Let Us Sum Up
- **4.8** Key Words
- **4.9** Further Readings and References

BLOCK 4: PREDICATE LOGIC

Unit 1: Introducing the Quantifiers 'All' and 'Some' and their Symbolic Representation

- 1.0 Objectives
- 1.1 Introduction
- 1.2 Symbolization of Propositions
- 1.3 Logical Relations involving Quantifiers
- 1.4 Fall-out of Universal and Existential Quantifiers
- 1.5 Examples
- 1.6 Exercises
- 1.7 Let Us Sum Up
- 1.8 Key Words
- 1.9 Further Readings and References

Unit 2: Rules of Universal Instantiation and Generalization, Existential Instantiation and Generalization, and Rules of Quantifier Equivalence

- 2.0 Objectives
- 2.1 Introduction
- 2.2 Rules of Quantification
- 2.3 Rules of Quantifier Equivalence
- 2.4 Application of the Quantification Rules
- 2.5 Examples
- 2.6 Exercises
- 2.7 Quantification Rules and Arguments
- 2.8 Let Us Sum Up
- 2.9 Key Words
- 2.10 Further Readings and References

Unit 3: Proofs of Validity

- **3.0** Objectives
- **3.1** Introduction
- **3.2** Quantification and Equivalence Relation
- **3.3** Rules of Quantification and Non-syllogism
- **3.4** Exercises
- **3.5** Multiply General Propositions
- **3.6** The Strengthened Rule of C. P. and Quantification
- **3.7** Let Us Sum Up
- **3.8** Key Words
- **3.9** Further Readings and References

Unit 4: Proving Invalidity

- **4.0** Objectives
- **4.1** Introduction
- **4.2** Methods of Proving Invalidity-1
- **4.3** Methods of proving Invalidity-2
- **4.4** Exercises
- **4.5** Let Us Sum Up
- **4.6** Key Words
- **4.7** Further Readings and References

Unit 5: Applications of Symbolic Logic

- **5.0** Objectives
- **5.1** Introduction
- **5.2** Application of Symbolic Logic with Digital Logic
- **5.3** Boolean Algebra
- **5.4** Logic Gates
- **5.5** Role of symbolic logic in Multi-Value logic
- **5.6** Application of Fuzzy Logic
- **5.7** Let Us Sum Up
- **5.8** Key Words
- **5.9** Further Readings and References